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ashishkumar-source add

75ad024 · 35 minutes ago



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We will be walking thourgh basic linux terminal commands

1. Navigation Commands

pwd – Print Working Directory

Shows the current location in the filesystem.

```
pwd
```



Output example:

```
/home/ashishkumar/Assignment
```

```
ashishkumar@ashishkumar:~/Assignment $ pwd
/home/ashishkumar/Assignment
```

ls – List Directory Contents

The ls command is used to list files and directories in the current working directory. flag-a list down all the file and folder including the one which are hidden



```
ls -l → Detailed list (permissions, size, date)
ls -a → Shows hidden files (those starting with .)
ls -la → Combined
```

output is :

```
ashishkumar@ashishkumar:~/Assignment $ ls
img10.png  img14.png  img17.png  'Screenshot from 2025-09-05 10-49-55.png'  'Screenshot from 2025-09-07 16-35-05.png'
img11.png  'img15(1).png'  img18.png  'Screenshot from 2025-09-05 10-51-20.png'
img12.png  img16-1.png  lab0.md    'Screenshot from 2025-09-05 10-51-33.png'
img13.png  img16.png  lab1.md    'Screenshot from 2025-09-05 10-52-12.png'

ashishkumar@ashishkumar:~/Assignment $ ls -a
.          img12.png  img16-1.png  lab0.md          'Screenshot from 2025-09-05 10-51-33.png'
..         img13.png  img16.png    lab1.md          'Screenshot from 2025-09-05 10-52-12.png'
img10.png  img14.png  img17.png    'Screenshot from 2025-09-05 10-49-55.png'  'Screenshot from 2025-09-07 16-35-05.png'
img11.png  'img15(1).png'  img18.png    'Screenshot from 2025-09-05 10-51-20.png'

ashishkumar@ashishkumar:~/Assignment $ ls -la
total 2644
drwxrwxr-x  2 ashishkumar ashishkumar  4096 Sep  7 16:35 .
drwxr-x--- 28 ashishkumar ashishkumar  4096 Sep  7 16:31 ..
-rw-rw-r--  1 ashishkumar ashishkumar 153136 Sep  6 18:50 img10.png
-rw-rw-r--  1 ashishkumar ashishkumar 247549 Sep  6 18:40 img11.png
-rw-rw-r--  1 ashishkumar ashishkumar 198174 Sep  6 18:51 img12.png
-rw-rw-r--  1 ashishkumar ashishkumar 148374 Sep  6 18:52 img13.png
-rw-rw-r--  1 ashishkumar ashishkumar 154310 Sep  6 18:57 img14.png
-rw-rw-r--  1 ashishkumar ashishkumar 148812 Sep  6 18:53 'img15(1).png'
-rw-rw-r--  1 ashishkumar ashishkumar 176663 Sep  6 18:55 img16-1.png
-rw-rw-r--  1 ashishkumar ashishkumar 176663 Sep  6 18:55 img16.png
-rw-rw-r--  1 ashishkumar ashishkumar 101645 Sep  6 18:58 img17.png
-rw-rw-r--  1 ashishkumar ashishkumar 281239 Sep  6 19:01 img18.png
-rw-rw-r--  1 ashishkumar ashishkumar  2668 Sep  6 19:54 lab0.md
-rw-rw-r--  1 ashishkumar ashishkumar    0 Sep  7 16:28 lab1.md
-rw-rw-r--  1 ashishkumar ashishkumar 248461 Sep  6 18:46 'Screenshot from 2025-09-05 10-49-55.png'
-rw-rw-r--  1 ashishkumar ashishkumar 242590 Sep  6 18:42 'Screenshot from 2025-09-05 10-51-20.png'
-rw-rw-r--  1 ashishkumar ashishkumar 124343 Sep  6 18:45 'Screenshot from 2025-09-05 10-51-33.png'
-rw-rw-r--  1 ashishkumar ashishkumar 250291 Sep  5 11:01 'Screenshot from 2025-09-05 10-52-12.png'
-rw-rw-r--  1 ashishkumar ashishkumar   6415 Sep  7 16:35 'Screenshot from 2025-09-07 16-35-05.png'
```

cd – Change Directory

Moves into a directory.

```
cd
```



Examples:

```
cd Documents      # Go to Documents
cd ..             # Go up one level
cd /              # Go to root
cd ~              # Go to home directory
```



```
ashishkumar@ashishkumar:~$ cd UPES
ashishkumar@ashishkumar:~/UPES$ cd linux_lab
ashishkumar@ashishkumar:~/UPES/linux_lab$ cd ..
ashishkumar@ashishkumar:~/UPES$ cd ~
ashishkumar@ashishkumar:~$ cd UPES
ashishkumar@ashishkumar:~/UPES$
```



2. File and Directory Management

mkdir – Make Directory

Creates a new folder.

```
mkdir new_folder
```



output :

```
ashishkumar@ashishkumar:~/UPES$ mkdir lab
ashishkumar@ashishkumar:~/UPES$ cd lab
ashishkumar@ashishkumar:~/UPES/lab$
```

touch – Create File

Creates an empty file.

```
touch file.txt
```



output:

```
ashishkumar@ashishkumar:~/UPES/lab$ touch lab2.c
ashishkumar@ashishkumar:~/UPES/lab$ ls
lab1  lab2.c
```

cp – Copy Files or Directories

```
cp source.txt destination.txt
```



mv – Move or Rename Files

```
mv oldname.txt newname.txt
```



rm – Remove Files

```
rm file.txt          # Delete file
rm -r folder_name    # Delete folder (recursively)
```



Be careful! There is no undo.

```
ashishkumar@ashishkumar:~/UPES/lab$ rm lab2.c
ashishkumar@ashishkumar:~/UPES/lab$ ls
lab1
```



3. File Viewing & Editing

cat – View File Contents

Displays content in terminal.

```
cat file.txt
```



Output:

```
ashishkumar@ashishkumar:~/UPES/lab$ cat lab1
Hello
My Name is Ashish Choudhary.
```

nano – Edit Files in Terminal

A basic terminal-based text editor.

```
nano file.txt
```



Use arrows to move
CTRL + O to save
CTRL + X to exit



output :

```
ashishkumar@ashishkumar:~/UPES/lab$ nano lab1
```

clear – Clears the Terminal

clear



Shortcut: CTRL + L

✓ 4. System Commands

echo – Print Text

Useful for debugging or scripting.

```
echo "Hello, World!"
```



output :

```
ashishkumar@ashishkumar:~/UPES/lab$ echo "hello "  
hello
```

whoami – Show Current User

whoami



output :

```
ashishkumar@ashishkumar:~/UPES/lab$ whoami  
ashishkumar
```

man – Manual for Any Command

```
man ls
```



Use `q` to quit the manual.

✓ 5. Searching and Finding

find – Locate Files

```
find . -name "*.txt"
```



 Finds all `.txt` files in current folder and subfolders.

output :

```
ashishkumar@ashishkumar:~/Desktop$ find odd.c
odd.c
```

grep – Search Inside Files

```
grep "hello" file.txt
```



```
ashishkumar@ashishkumar:~/Desktop$ grep "int" odd.c
int main (){
for(int i=5;i<=50;i++){
printf("%d\n",i);
ashishkumar@ashishkumar:~/Desktop$
```

 Searches for the word `hello` inside `file.txt` .

✓ 6. Helpful Shortcuts

Shortcut	Action
Tab	Auto-complete files/folders

Shortcut	Action
↑ / ↓	Browse command history
CTRL + C	Stop a running command
CTRL + L	Clear screen

✓ 7. Bonus: Chaining Commands

- Run multiple commands:

```
mkdir test && cd test && touch hello.txt
```



- Run only if previous command succeeds: &&
- Run regardless of success: ;

Shell Tutorial – File Permissions with chmod and chown

◆ 1. Understanding File Permissions in Linux

Each file/directory in Linux has:

- **Owner** → The user who created the file.
- **Group** → A group of users who may share access.
- **Others** → Everyone else.

Permission Types

- **r** → Read (4 in numeric)
- **w** → Write (2 in numeric)
- **x** → Execute (1 in numeric)

Permission Layout

Example from `ls -l` :

-rwxr-xr--



Breakdown:

- - → Regular file (d = directory, l = symlink, etc.)
- rwx → Owner has read, write, execute
- r-x → Group has read, execute
- r-- → Others have read only

output :

```
ashishkumar@ashishkumar:~$ ls -la
total 216
drwxr-x--- 28 ashishkumar ashishkumar 4096 Sep  7 18:06 .
drwxr-xr-x  3 root        root        4096 Aug  6 23:23 ..
-rwxrwxr-x  1 ashishkumar ashishkumar 16000 Aug 15 00:25 a.out
drwxrwxr-x  2 ashishkumar ashishkumar 4096 Sep  4 23:00 ashish
drwxrwxr-x  2 ashishkumar ashishkumar 4096 Sep  7 17:19 'Assignment '
-rw-----  1 ashishkumar ashishkumar 8443 Sep  7 18:06 .bash_history
-rw-r--r--  1 ashishkumar ashishkumar  220 Aug  6 23:23 .bash_logout
-rw-r--r--  1 ashishkumar ashishkumar 3771 Aug  6 23:23 .bashrc
drwx----- 17 ashishkumar ashishkumar 4096 Sep  7 18:06 .cache
drwxrwxr-x 19 ashishkumar ashishkumar 4096 Sep  7 21:23 .config
drwxr-xr-x  7 ashishkumar ashishkumar 4096 Sep  1 23:00 Desktop
drwxr-xr-x  2 ashishkumar ashishkumar 4096 Aug  6 23:24 Documents
drwxrwxr-x  3 ashishkumar ashishkumar 4096 Aug  8 18:28 .dotnet
drwxr-xr-x  6 ashishkumar ashishkumar 4096 Sep  7 16:32 Downloads
-rwxrwxrwx  1 ashishkumar ashishkumar  784 Aug 30 13:46 ds_starter
-rw-r--r--  1 ashishkumar ashishkumar 4096 Aug 30 13:51 ds_starter.db
-rw-----  1 ashishkumar ashishkumar    7 Sep  7 18:06 file.txt.save
drwxrwxr-x  8 ashishkumar ashishkumar 4096 Sep  7 17:23 .git
-rw-rw-r--  1 ashishkumar ashishkumar   69 Aug  9 12:41 .gitconfig
drwx-----  2 ashishkumar ashishkumar 4096 Aug 12 13:03 .gnupg
-rw-----  1 ashishkumar ashishkumar   20 Aug 21 23:17 .lessht
drwxrwxr-x  3 ashishkumar ashishkumar 4096 Aug 13 12:29 linuxlab
drwxrwxr-x  4 ashishkumar ashishkumar 4096 Aug  6 23:24 .local
-rw-rw-r--  1 ashishkumar ashishkumar 24664 Sep  6 22:14 mozilla.pdf
drwxr-xr-x  2 ashishkumar ashishkumar 4096 Aug  8 11:07 Music
drwxrwxr-x  2 ashishkumar ashishkumar 4096 Sep  7 16:31 new_folder
-rw-r--r--  1 ashishkumar ashishkumar  361 Aug  6 23:23 .pam_environment
drwxr-xr-x  3 ashishkumar ashishkumar 4096 Aug  8 17:38 Pictures
drwx-----  3 ashishkumar ashishkumar 4096 Aug  8 17:38 .pki
-rw-r--r--  1 ashishkumar ashishkumar  807 Aug  6 23:23 .profile
drwxr-xr-x  2 ashishkumar ashishkumar 4096 Aug  6 23:24 Public
-rw-rw-r--  1 ashishkumar ashishkumar   24 Sep  5 15:42 README.md
-rw-rw-r--  1 ashishkumar ashishkumar  727 Aug 30 13:51 requirements.txt
-rw-rw-r--  1 ashishkumar ashishkumar  145 Aug 27 13:02 second.sh
drwx-----  7 ashishkumar ashishkumar 4096 Aug  8 18:11 snap
drwx-----  2 ashishkumar ashishkumar 4096 Aug 12 12:10 .ssh
```


◆ 2. chmod – Change File Permissions

Syntax

```
chmod [options] mode filename
```



Modes can be set in **numeric (octal)** or **symbolic** form.

(A) Numeric (Octal) Method

Each permission is represented as a number:

- Read = 4
- Write = 2
- Execute = 1

Add them up:

- 7 = rwx
- 6 = rw-
- 5 = r-x
- 4 = r--
- 0 = ---

Example:

```
chmod 777 script.sh
```



Meaning:

- Owner: 7 → rwx
- Group: 7 → r-w-x
- Others: 7 → r-w-x

image

```
ashishkumar@ashishkumar:~/UPES/linux_lab$ chmod 000 experiment4.md/
d----- 3 ashishkumar ashishkumar 4096 Aug 22 11:21 experiment4.md
ashishkumar@ashishkumar:~/UPES/linux_lab$ chmod 777 experiment4.md
drwxrwxrwx 3 ashishkumar ashishkumar 4096 Aug 22 11:21 experiment4.md
```

(B) Symbolic Method

Use u (user/owner), g (group), o (others), a (all). Operators:

- + → Add permission
- - → Remove permission
- = → Assign exact permission

Modes can be set in **numeric (octal)** or **symbolic** form.

output:

```
ashishkumar@ashishkumar:~/UPES$ chmod a-r file.txt
ashishkumar@ashishkumar:~/UPES$ ls -la
total 16
drwxrwxr-x 4 ashishkumar ashishkumar 4096 Sep  7 22:25 .
drwxr-x--- 28 ashishkumar ashishkumar 4096 Sep  7 18:06 ..
-rw-rw-r-- 1 ashishkumar ashishkumar  0 Aug 22 10:54 d.txt
--w--w---- 1 ashishkumar ashishkumar  0 Sep  7 22:25 file.txt
```

(C) Recursive Changes

```
chmod -R 755 /mydir
```



- -R → applies changes recursively to all files/subdirectories.

```
ashishkumar@ashishkumar:~/UPES$ chmod 755 file.txt
ashishkumar@ashishkumar:~/UPES$ ls -la
total 16
drwxrwxr-x 4 ashishkumar ashishkumar 4096 Sep  7 22:25 .
drwxr-x--- 28 ashishkumar ashishkumar 4096 Sep  7 18:06 ..
-rw-rw-r-- 1 ashishkumar ashishkumar  0 Aug 22 10:54 d.txt
-rwxr-xr-x 1 ashishkumar ashishkumar  0 Sep  7 22:25 file.txt
```

```
ashishkumar@ashishkumar:~/UPES$ mkdir parent
ashishkumar@ashishkumar:~/UPES$ cd parent/
ashishkumar@ashishkumar:~/UPES/parent$ mkdir child
ashishkumar@ashishkumar:~/UPES/parent$ cd child/
ashishkumar@ashishkumar:~/UPES/parent/child$ touch test.txt
ashishkumar@ashishkumar:~/UPES/parent/child$ ls -la
total 8
drwxrwxr-x 2 ashishkumar ashishkumar 4096 Sep  7 22:35 .
drwxrwxr-x 3 ashishkumar ashishkumar 4096 Sep  7 22:34 ..
-rw-rw-r-- 1 ashishkumar ashishkumar    0 Sep  7 22:35 test.txt
ashishkumar@ashishkumar:~/UPES/parent/child$ chmod 000 test.txt
ashishkumar@ashishkumar:~/UPES/parent/child$ ls -la
total 8
drwxrwxr-x 2 ashishkumar ashishkumar 4096 Sep  7 22:35 .
drwxrwxr-x 3 ashishkumar ashishkumar 4096 Sep  7 22:34 ..
----- 1 ashishkumar ashishkumar    0 Sep  7 22:35 test.txt
ashishkumar@ashishkumar:~/UPES/parent/child$ cd .
./ ../
ashishkumar@ashishkumar:~/UPES/parent/child$ cd ..
ashishkumar@ashishkumar:~/UPES/parent$ chmod 000 child/
ashishkumar@ashishkumar:~/UPES/parent$ ls -la
total 12
drwxrwxr-x 3 ashishkumar ashishkumar 4096 Sep  7 22:34 .
drwxrwxr-x 5 ashishkumar ashishkumar 4096 Sep  7 22:34 ..
d----- 2 ashishkumar ashishkumar 4096 Sep  7 22:35 child
ashishkumar@ashishkumar:~/UPES/parent$ cd ..
ashishkumar@ashishkumar:~/UPES$ chmod 000 parent/
ashishkumar@ashishkumar:~/UPES$ ls -ls
total 12
4 drwxrwxr-x  2 ashishkumar ashishkumar 4096 Sep  7 17:03 lab
4 drwxrwxr-x 10 ashishkumar ashishkumar 4096 Sep  3 23:27 linux_lab
4 d-----  3 ashishkumar ashishkumar 4096 Sep  7 22:34 parent
ashishkumar@ashishkumar:~/UPES$ chmod 777 parent/
ashishkumar@ashishkumar:~/UPES$ ls -la
total 20
drwxrwxr-x  5 ashishkumar ashishkumar 4096 Sep  7 22:34 .
drwxr-x--- 28 ashishkumar ashishkumar 4096 Sep  7 18:06 ..
drwxrwxr-x  2 ashishkumar ashishkumar 4096 Sep  7 17:03 lab
drwxrwxr-x 10 ashishkumar ashishkumar 4096 Sep  3 23:27 linux_lab
drwxrwxrwx  3 ashishkumar ashishkumar 4096 Sep  7 22:34 parent
```

```
ashishkumar@ashishkumar:~/UPES$ chmod -R 777 parent/
ashishkumar@ashishkumar:~/UPES$ ls -la
total 20
drwxrwxr-x  5 ashishkumar ashishkumar 4096 Sep  7 22:34 .
drwxr-x--- 28 ashishkumar ashishkumar 4096 Sep  7 18:06 ..
drwxrwxr-x  2 ashishkumar ashishkumar 4096 Sep  7 17:03 lab
drwxrwxr-x 10 ashishkumar ashishkumar 4096 Sep  3 23:27 linux_lab
drwxrwxrwx  3 ashishkumar ashishkumar 4096 Sep  7 22:34 parent
ashishkumar@ashishkumar:~/UPES$ cd parent/
ashishkumar@ashishkumar:~/UPES/parent$ ls -la
total 12
drwxrwxrwx 3 ashishkumar ashishkumar 4096 Sep  7 22:34 .
drwxrwxr-x 5 ashishkumar ashishkumar 4096 Sep  7 22:34 ..
drwxrwxrwx 2 ashishkumar ashishkumar 4096 Sep  7 22:35 child
ashishkumar@ashishkumar:~/UPES/parent$ cd child/
ashishkumar@ashishkumar:~/UPES/parent/child$ ls -la
total 8
drwxrwxrwx 2 ashishkumar ashishkumar 4096 Sep  7 22:35 .
drwxrwxrwx 3 ashishkumar ashishkumar 4096 Sep  7 22:34 ..
-rwxrwxrwx 1 ashishkumar ashishkumar   0 Sep  7 22:35 test.txt
```

◆ 3. chown – Change File Ownership

Syntax

```
chown [options] new_owner:new_group filename
```



Examples:

```
chown ashish.txt           # Change owner to user 'sameer'
chown ashish:dev.txt        # Change owner to 'sameer' and group to
chown ashish:dev file.txt   # Change only group to 'dev'
chown -R ashish:dev /project # Recursive ownership change
```



Practice Experiment on chown

◆ 1. Create a new user

```
sudo useradd -m newuser
```



- -m → creates a home directory /home/newuser .

♦ 2. Create a new group

```
sudo groupadd newgroup
```



♦ 3. Add the user to the group

```
sudo usermod -aG newgroup newuser
```



- -aG → append user to the supplementary group (doesn't remove existing groups).

♦ 4. Create a file (as current user, e.g. root or your login user)

```
touch testfile.txt
```



Check ownership:

```
ls -l testfile.txt
```



Example:

```
-rw-rw-r--1 sameerchoudhary sameerchoudhary 0 Aug 20 18:52  
testfile.txt
```



♦ 5. Assign ownership of the file to newuser and newgroup

```
sudo chown newuser:newgroup testfile.txt
```



♦ 6. Verify ownership

```
ls -l testfile.txt
```



Output:

```
-rw-rw-r-- 1 newuser newgroup 0 Aug 20 18:52 testfile.txt
```



✅ **Key Tip:** Use **numeric** for quick settings (e.g., 755, 644) and **symbolic** for fine adjustments (`u+x` , `g-w`).

```
ashishkumar@ashishkumar:~/UPES/parent$ sudo useradd -m newuser
ashishkumar@ashishkumar:~/UPES/parent$ sudo groupadd newgroup
groupadd: group 'newgroup' already exists
ashishkumar@ashishkumar:~/UPES/parent$ sudo usermod -aG newgroup newuser
ashishkumar@ashishkumar:~/UPES/parent$ touch testfile.txt
ashishkumar@ashishkumar:~/UPES/parent$ ls -l testfile.txt
-rw-rw-r-- 1 ashishkumar ashishkumar 0 Sep  7 22:56 testfile.txt
ashishkumar@ashishkumar:~/UPES/parent$ sudo chown newuser:newgroup testfile.txt
ashishkumar@ashishkumar:~/UPES/parent$ ls -l testfile.txt
-rw-rw-r-- 1 newuser newgroup 0 Sep  7 22:56 testfile.txt
ashishkumar@ashishkumar:~/UPES/parent$
```

◆ 4. Putting It All Together

Example Scenario

```
touch project.sh
ls -l project.sh
```



Output:

```
-rw-r--r-- 1 sameer dev 0 Aug 19 12:00 project.sh
```



Now:

```
chmod 700 project.sh          # Only owner has rwx
chmod u+x,g-w project.sh     # Add execute for user, remove write for group
chown root:admin project.sh  # Change owner to root and group to admin
```



◆ 5. Quick Reference Table

Numeric	Permission	Meaning
0	---	No access
1	--x	Execute only
2	-w-	Write only
3	-wx	Write + Exec
4	r--	Read only
5	r-x	Read + Exec
6	rw-	Read + Write
7	rwX	Full access

Q1 what is the difference between chmod and chown?

ANS=chown-change ownership

change the owner and group of a file or directory



chmod-change permissions

changes the permissions for the owner,group, and others

Q2 how do you check current directory and user?

ANS = by using the pwd command to check current directory and
by using the whoami to check current user.

